



Career Options in Sciences

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Physical Sciences

Introduction

Physical science is the study of physics and chemistry of nature. It overlaps life sciences in a way to include ecology and the evidence of historical facts of evolution.

Eligibility

- 10+2 or its equivalent with Mathematics, Physics, Chemistry
- For IIT's, it is mandatory to qualify in the Joint Entrance Examinations (J.E.E)

Courses

- B.Sc / M.Sc
- B.Tech /M.Tech Biological Engineering

Institutes/Universities

- Indian Institute of Technology, New Delhi
- Indian Institute of Technology, Madras
- Indian Institute of Technology, Guwahati
- Indian Institute of Space Science and Technology, Thiruvananthapuram
- Indian Institute of Technology, Bombay
- National Institute of Technology, Calicut
- Technology Education & Research Integrated Institutions Kurukshetra
- Guru Gobind Singh Indraprastha University, New Delhi

Astronomy and Astrophysics

Introduction

Astronomy is a combination of physics, chemistry and mathematical principles/rules. Astrophysics can be called its offshoot. It deals with detailed study of the physical, chemical and dynamic properties of celestial objects. It also deals with the phenomena over and above Earth's atmosphere. There is associated study of calculations of orbits, gravitational forces, satellites, meteors, galaxies, comets, stars, planetary objects, planets, satellites etc. In Astrophysics, we explore and ensure properties/nature of the astronomical objects with the help of laws of physics and chemistry. There is also the field of Cosmology which studies the origin and evolution of the universe.

Eligibility

10 +2 with PCM

Entrance Tests (PhD from IUCAA):

- IUCAA-NCRA Admission Test (INAT) Link Details:
<http://www.ncra.tifr.res.in/ncra/students/external-students/inat>
- Joint Entrance Screening Test (JEST) Details: <http://www.jest.org.in/>
- CSIR-UGC NET for JRF (Physics)

Courses

- M.Sc. /M.Phil PhD (Physics)
- M.Sc. Astronomy
- M.Sc. Astrophysics

- Integrated M.Tech- Ph. D (Tech.) programme in Astronomical Instrumentation (Eligibility - B. Tech / BE degree in Electrical/ Instrumentation/ Electronics and Communications /Computer Science/Mechanical Engineering or M. Sc degree in Physics / Electronic Science / Astrophysics/ Applied Mathematics /Applied Physics are also eligible to apply)
- Ph.D Astrophysics / Astronomy

Institutes/Universities

- University of Delhi, Delhi
- Indian Institute of Astrophysics, Bangalore
- Indian Institute of Science, Bangalore
- Raman Research Institute, Bangalore
- Inter-University Centre for Astronomy and Astrophysics (IUCAA) – Pune
- National Centre for Radio Astronomy - Tata Fundamental Research Institute – Pune
- Aryabhata Research Institute of Observational Sciences (ARIES), Nainital
- Harish-Chandra Research Institute (HRI), Allahabad
- Osmania University, Hyderabad
- Madras University, Chennai

Agricultural Science

Introduction

Agricultural Science is the study of production of food and other goods i.e. plants and living organisms in a systematic and controlled manner. This field incorporates management of farms, horticulture and agri-business activities. It also gives insights of agrarian scenario of the country and its agriculture industry that manufacture agricultural machinery and equipment and is involved in procurement and processing of agri-products. It also studies banking activities for financing and developing farms, research and development for the purpose of improving capacity and quality of farm products and so on.

Agricultural science students can specialize in agricultural production and livestock.

Eligibility

10+2 or equivalent having passed with Physics, Chemistry, Biology/Math or Agriculture stream

Certificate/Diploma Courses

Eligibility - A pass in the Higher Secondary Examination (10+2) or equivalent with related subjects

- Diploma in Agriculture
- Diploma in Horticulture

Institutes/Universities

- CCS Agriculture University, Hisar, Haryana
- University of Agricultural Sciences, Bangalore
- Acharya N G Ranga Agricultural University, Guntur, [Andhra Pradesh](#)
- Andhra University, Visakhapatnam
- Kerala Agricultural University (KAU), Thrissur, Kerala
- [Tamil Nadu](#) Agricultural University, Coimbatore
- Dr R Prasad Central Agriculture University, Samastipur, Bihar

UG/PG Courses

- Bachelor of Science (Honours) in Agriculture
- Bachelor of Fisheries Science (B.F.Sc.)
- B. Sc. Food Technology
- B. Sc. Horticulture
- B. Tech Biotechnology
- B. Tech. Agriculture Engineering
- Master of Science in Agriculture
- MBA Agri-business
- M.Sc. Agricultural Biotechnology
- Master of Science in Agriculture
- Doctor of Philosophy in Agriculture

Biological Science

Introduction

Biological Science studies life and living organisms, their life cycles, adaptations and environment. There are various fields of study under the biological sciences including biochemistry, microbiology and evolutionary biology.

Eligibility

10+2 with Physics, Chemistry and Biology as compulsory subjects

Courses

- Bachelor in Biological Science
- Master in Biological Science
- PhD in Biological Science

Institutes/Universities

- Indian Institute of Science Education and Research (IISER), Pune, Maharashtra.
- Indian Institute of Science Education and Research (IISER), Kolkata, [West Bengal](#).

Biotechnology

Introduction

Biotechnology is utilizing the sciences of biology, chemistry, physics, engineering and information technology to develop tools and products to apply to living cells/organisms.

Genetic Engineering

Genetic engineering is the study of knowledge obtained from genetics to alter the reproduction and hereditary processes of organisms. It deals in cloning, in-vitro fertilization, species hybridization or direct manipulation of the genetic material itself by the recombinant DNA technique.

Plant Tissue Culture

Functions in the field of in-vitro regeneration and propagation of plantlets, a Tissue Culture specialist may also be involved in plant germ plasma conservation.

Plant Genetics

Activities to improve plants by evolving new horticultural varieties through hybridization, this work is generally carried out in the laboratories.

Eligibility & Courses

- For B. Sc. Courses: Pass in 10+2 with PCM. (Biology)
- For M. Sc. Courses: Bachelor's degree under 10+2+3 in Physical, Biological, Agricultural, Veterinary, Fishery Sciences, Pharmacy, Engineering, Technology or Medicine (MBBS)
- For M. Sc. (Agriculture) Biotechnology/M.Sc. Animal Biotechnology: Bachelor's degree in agriculture, horticulture, forestry, fishery, Veterinary Sciences or Agricultural Engineering.
- For M Tech Biotechnology : B. Tech degree - in chemical Engineering, Biochemical Engineering, Industrial Biotech (BE), Leather technology, Pharmaceutical Technology, Food Technology, B. Pharma, and Dairy Technology or Master's Botany, Zoology Bio-chemistry, Microbiology, Genetics, Physiology, Pharmacology and Biophysics.

All India Biotechnology Entrance Examination

Jawaharlal Nehru University, New Delhi conducts the Combined Entrance examination for admission to M.Sc./M.Sc. (Agri.)/M.V.Sc.(Animal) Biotech/M. Tech Biotechnology in association with other universities in India. Candidates who have done their B. Sc. in physical, biological, agricultural, veterinary and fishery sciences, pharmacy, engineering, technology or medicine can take up this examination.

Education Abroad

The Ministry of Science and Technology awards the Biotechnology Overseas Associateship for advanced research in molecular biology, microbial genetics, gene therapy, virology, tissue culture and so on.

Computer Applications

Introduction

The Computer Applications course is designed to provide the student with the opportunity to expand and apply technological knowledge.

Eligibility

10+2 or equivalent

Courses

- Diploma in Computer Application
- Bachelor in Computer Application
- Master in Computer Application

Institutes/Universities

- University of Madras, Chennai
- Chhatrapati Shahuji Maharaj University, Kanpur,
- University of Allahabad, Allahabad
- University of Mumbai, Mumbai
- National Institute of Electronics & Information Technology, New Delhi
- Delhi University, Delhi

Note: Computer Application Courses are also offered by various private institutions. Therefore, aspirants are advised to ensure the recognition and approved courses prior to admission.

Computer Science

Introduction

Computer science is one of the known courses among engineering aspirants which aims on the basic elements of computer programming and networking. This course includes knowledge of design, implementation and management of information system of both hardware and software. It deals principally with the theory of computation and design of computational systems.

Eligibility

10+2 with (PCM) and followed by Entrance Test

Courses offered by NIELT (DOEACC)

- A level – Advanced diploma: equal to Bachelor's degree
- B level – equal to MCA degree 3. C level – equal o M. Tech

Courses

UG Courses

- B.E. / B. Tech (Comp. Science, Info. Science, Info Tech)
- B. Sc. (CS, IT, IS)
- BCA.

PG Courses

- ME / M. Tech
- MCA.
- M. Sc. (CS, IT)
- PGDIT, PGDCA, etc.

Doctoral Courses

- Ph. D

Institutes/Universities

- Indian Institute of Technology Bombay, Mumbai
- Indian Institute of Technology Delhi
- Indian Institute of Technology Kanpur
- Indian Institute of Technology Kharagpur
- Delhi Technological University, Delhi
- Banaras Hindu University, Varanasi

- Indian School of Mines Dhanbad, Jharkhand
- [Indira Gandhi National Open University](https://www.ignou.ac.in/), New Delhi (<https://www.ignou.ac.in/>)

Earth Sciences/Geography

Introduction

Earth Sciences deals with study and research in various fields and concepts of Geology, examination and utilization of natural resources their management, sustenance and conservation.

Earth Sciences/Geography, studies climatic/ weather conditions of different regions with periodic developments, their effects, natural resources like water, minerals survey, earth and ocean characteristics and its science, air, rain, clouds and rivers aspects, monitoring of geographical environments, forests and weather forecast. The special study are in this field like Economic & Political Geography, Human Geography, Cultural Geography, Environment and Natural Resources Studies. The other aspects of Geography include Geomorphology, Climatology, Oceanography, Meteorology, Cartography, Geographic Information Systems and other allied areas.

Eligibility

10+2 or equivalent

Courses

- BA with Geography as a subject
- BA Honors (Geography)
- Integrated M. Tech. (Geophysical Technology)
- Integrated M. Tech. (Geological Technology)
- M. Tech. Earth Sciences
- M. Sc. (Applied Geology)
- M. Phil. Geology
- M. Phil. Geoinformatics
- Ph.D. Geology
- MA Geography
- M. Sc Geography
- M. Sc., M.Phil. & Ph. D
- M. Sc. Geology (Five Year Integrated)
- M. Sc. Geology (Two Year CBCS)
- PG Diploma in Petroleum Geoscience
- PG Diploma in Remote Sensing and GIS

Institutes /Universities

- Indian Institute of Technology, Roorkee, Uttarakhand
- Indian Institute of Science, Bengaluru
- Bangalore University, Bengaluru
- Kurukshetra University, Kurukshetra, Haryana,
- University of Delhi
- MD University, Rohtak, Haryana
- Annamalai University, Annamalainagar, Tamil Nadu
- Indira Gandhi National Open University, New Delhi (<https://www.ignou.ac.in/>)

Environmental Sciences

Introduction

Environmental science is the area of study that deals with the interactions of the physical, chemical, and biological elements of the environment and impact of these upon the living beings and the environment. Generally, environmental science is considered to be only natural science whereas it includes the study of social science and humanities also. Therefore, it is an interdisciplinary course.

Eligibility

10+2 examination or equivalent with Physics, Chemistry and Biology.

Courses

- Bachelor's Degree
- Master's Degree
- Ph. D (Environmental Science) 10+2

Institutes/Universities

- Allahabad University, Allahabad, [Uttar Pradesh](#).
- Delhi University, Delhi
- School of Environmental Sciences, Jawaharlal Nehru University
- Babasaheb Bhimrao Ambedkar University, Lucknow, Uttar Pradesh.
- Banaras Hindu University, Varanasi, Uttar Pradesh.
- Bangalore University, Bengaluru.
- Barkatullah University, Bhopal.
- Central University of Haryana, Mahendragarh,
- Central University of Jammu, Samba
- Central University of Jharkhand, Ranchi
- Dr. B. R. Ambedkar Open University, Telangana

Fishery Sciences

Introduction

Fisheries is a multidisciplinary course that includes the biological study of life, habits and breeding of various species of fish, It involves catching, processing, marketing and conservation of fish.

Eligibility

10+2 with biological science as one of the subjects.

Courses

- Bachelor of Fishery Science
- Bachelor of Science in Fisheries
- Bachelor of Fisheries Science in Fish Biology
- Bachelor of Fisheries Science in Aquaculture
- Master of Fisheries Science

Apart from the B.F.Sc & M.F.Sc. various fisheries institutes offer vocational training courses related to fisheries and allied disciplines.

Institutes/Universities

- Andhra University
- Barkatullah University, Bhopal, [Madhya Pradesh](#).
- G.B. Pant University of Agriculture, Pant Nagar, Uttaranchal.
- Goa University,
- Gujarat Agricultural University,
- Imphal Fisheries Training College, Manipur.
- Manipur University,
- Marathwada University, Aurangabad, Maharashtra.
- Osmania University, Hyderabad, Andhra Pradesh.

Floriculture / Horticulture

Introduction

Horticulture/Floriculture is one of the parts of agriculture that study the art, science, technology, and process of growing plants. It involves fruits, vegetables, nuts, seeds, herbs, sprouts, mushrooms, algae, flowers, the cultivation of medicinal plants, seaweeds and non-food crops such as grass and ornamental trees and plants.

Eligibility

10+2 with physics, chemistry and maths/biology/ agriculture

10+2 with physics, chemistry and mathematics/ agriculture.

Courses

- Diploma in Horticulture
- B.Sc. in Horticulture
- M. Sc. in Horticulture
- Ph. D in Horticulture

Institutes/Universities

- Indian Agricultural Research Institute (IARI), Delhi.
- Kerala Agricultural University
- University of Agricultural Sciences, Bengaluru.
- Allahabad Agricultural Institute, Allahabad
- Tamil Nadu Agricultural University, Coimbatore
- Calcutta University, Kolkata
- Acharya N.G. Ranga Agricultural University, Hyderabad.
- Chaudhary Charan Singh Haryana Agricultural University (CCSHAU), Hisar, Haryana.
- G B Pant University of Agriculture and Technology (GBPUA&T), Uttarakhand

Food Technology

Introduction

Food Technology is the study of food science in manufacturing safe, wholesome and nutritious food products. Training in Food technology gives adequate knowledge regarding the quality analysis of raw materials, packaging standards and methodology

Eligibility

10+2 education completed in the Science stream, from a recognized educational Board. Chemistry, Mathematics, Physics, Biology/ Home Science as the main subjects at the 10+2 level.

Courses

- Diploma Course
- Bachelor in Science
- Master in Science

Institutes/Universities

- Central Food Research Technological Research Institute, Karnataka
- University of Delhi
- Dr. Babasaheb Ambedkar Marathwada University, Aurangabad, Maharashtra

- University of Calcutta, West Bengal
- University of Madras , Chennai, Tamil Nadu
- Jadavpur University , Kolkata, West Bengal
- Andhra University , Visakhapatnam, Andhra Pradesh
- University of Bombay , Mumbai, Maharashtra
- Mahatma Gandhi University , Kottayam, Kerala
- Manipur University, Imphal, Manipur

Home Science

Introduction

Home Science course is the study of the artistic values and the scientific methods employed in home management. It helps the students to have knowledge on diverse aspects of Resource Management, Food Nutrition, Human development, and so on.

Eligibility

10+2 or intermediate examination or an equivalent examination with any two Biology/Natural Sciences, Physical Sciences, Agricultural Sciences and Vocational courses in Home Science.

Courses

- B. Sc. in Home Science
- M.Sc. Home Science
- Ph.D. in Food and Nutrition
- Ph.D. in HECM

Colleges/Institutes

- College of Home Science, Bikaner (Rajasthan)
- College of Home Science, Hyderabad (Andhra Pradesh)
- College of Home Science, Pantnagar, Uttarakhand.
- Goa College of Home Science, Goa
- Government Home Science College, Chandigarh

Forestry

Introduction

Forestry is the study of sustainably managing and preserving forested resources for the benefit of humans. It studies ecology, environmental and resource protection, such as wood, water, wastelands, endangered species, clean air, biodiversity and ecosystems in forests.

Eligibility

10+2 with Physics, Chemistry, Maths/Biology.

Courses

- B. Sc. Forestry / Wildlife
- M. Sc. Forestry / Wildlife / Forest Economics
- M. Sc. Wood Science and Technology.
- Post Graduate Diploma in Forest Management (PGDFM)

Institutes/Universities

- University of Agricultural Sciences, Bengaluru, Dharwad.
- Agriculture College & Research Institute, Coimbatore.
- College of Horticulture Forestry, Solan, [Himachal Pradesh](#).
- Bhim Rao Ambedkar University, Agra, Uttar Pradesh.
- Gujarat Agricultural University, Sardar Krushinagar Gujarat.
- Orissa University of Agriculture & Technology, Bhubaneswar, Odisha
- Punjab Agricultural University, Ludhiana, Punjab.
- Jawaharlal Nehru Krishi Vishwavidyalaya, Jabalpur, Maharashtra.

Oceanography

Introduction

The course in Oceanography deals with the physical and biological aspects of the ocean and covers the study of ecosystem, geological and geophysical aspects of oceans; ocean waves currents, plate tectonics etc.

Eligibility

10+2 with Physics, Chemistry, Maths/Biology and followed by B. Tech. or B.Sc. in Zoology/ Botany/ Chemistry/ Fishery Science / Earth Science/ Physics/ Agriculture/ Microbiology/ Applied Sciences or equivalent or related subjects.

Courses

- M. Sc. Oceanography and Coastal Area Studies.
- M. Sc. Oceanography
- M. Phil. Chemical Oceanography
- M. Tech Ocean Technology
- M. Sc. Marine Biology

Institutes/Universities

- Indian Institute of Information Technology Madras, Chennai, Tamil Nadu.
- Indian Institute of Information Technology Kharagpur, West Bengal.

- Alagappa University, Karaikudi, Tamil Nadu
- University of Science and Technology, Cochin, Kerala.
- Goa University,
- Mangalore University, Karnataka.

Liberal Studies

Introduction

Liberal studies involves the multidisciplinary study of subjects ranging from Humanities to, Sciences, Social Sciences and Arts. The aim of the course is to develop critical thinking.

Courses

- BA/BSc Liberal Arts
- MA Liberal Studies
- Doctor in Liberal Studies

Eligibility

- 10+2 or equivalent examination

Institutes/Universities

- PDP University, Gandhinagar
- University of Delhi
- Jawaharlal Lal Nehru University, Delhi

Library Sciences

Introduction

Library Science course is the study of managing, maintaining and preserving records and information. Information of various types is sourced and classified for users with the help of record management preserved and disseminated with the help of technology.

Courses

- Certificate Course in Library Science.
- Diploma in Library Science.
- Bachelor of Lib & Info Science (BLISc)
- Bachelor of Library science (BLSc)
- Masters in Lib & Info Science (BLISc)
- Masters in Librarv science (BLSc)

Eligibility

10+2 (any stream) for Certificate & Diploma Courses

Institutes /Universities

- Jai Narain Vyas University
- University Of Rajasthan, Rajasthan
- Nalanda Open University, Patna, Bihar.
- Medical Library Association of India (MLAI),
- AIIMS, New Delhi offers a diploma course in Medical Librarianship.
- Bachelor of Library & Information sciences
- U.P. Rajarshi Tandon Open University, Allahabad, U.P.
- Yashwantrao Chavan Maharashtra Open University, Nashik, Maharashtra. Kota Open University, Kota, Rajasthan.
- Indira Gandhi National Open University, New Delhi (<http://www.ignou.ac.in>)
- Dr.B.R. Ambedkar Open University, Telangana

Statistical Sciences

Introduction

Statistics is the study data and navigating common problems for drawing correct conclusions. This course is related to the field such as areas as financial markets, sports, engineering, healthcare, marketing & sales, election campaigns, space, natural disasters, population studies, accidents, insurance, and deaths – statistics. Statistics deals with interpretation and aggregation of large complex data into simpler data. It develops from the field of probability in mathematics. Statistics also include the fields like marketing, economics, biology, public health, sports, medicines

Eligibility

10+2 pass with Maths & English.

Courses Offered

- Diploma in Applied Maths.
- Diploma in Maths with computer programming, etc.
- Bachelor of Statistics (Hons)
- Bachelor of Mathematics (Hons)
- Bachelor's in Statistical Methods.
- Bachelor's in Applied Mathematics & Statistics, etc.
- Masters in Statistics
- Masters in Mathematics.

- M. Tech. in Quality & Reliability and operations research.
- Masters of Science (MS) in Quantitative Economics
- P.G. Dip in Applied Maths (Industrial maths)
- M. Phil (Mathematics)
- Ph. D in Mathematics

Institutes/Universities

- Chennai Mathematical Institute, Chennai.
- College of Engineering, Chennai.
- Indian Institute of Technology, Kharagpur
- Indian Statistical Institute, Kolkata (HQ)
- Osmania University, Hyderabad.
- Tata Institute of Fundamental Research, Mumbai.
- The Institute of Mathematical Sciences (IMSC), Chennai.
- The Mehta Research Institute of Mathematics & Mathematical Physics (MRI), Allahabad.
- University of Hyderabad, Hyderabad.
- SavitriBai Phule Pune University

Source: [Central Board of Secondary Education, India](#)

source:

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